LUCAS PESENTI

POSITIONS AND EDUCATION

Scientific assistant since Sep. 2025
ETH Zürich

- Hosted by David Steurer

Ph.D. in Statistics and Computer Science Sep. 2021 – Jan. 2026
Bocconi University, Milan

- Thesis: "Algorithms beyond the union bound: polynomial optimization and discrepancy theory"
- Advised by Luca Trevisan (until June 2024), Laura Sanità and Pravesh Kothari (since June 2024).
- Awarded cum laude on January 21st, 2026.

B.Sc. and M.Sc. in Computer Science Sep. 2016 – Sep. 2021
École Normale Supérieure de Paris (ENS Ulm)

- Parisian Master of Research in Computer Science obtained with the highest honors (rank 1/65).
- Student and civil servant at ENS with a major in computer science and a minor in mathematics.
- Ranked 1st at the ENS entrance exam (computer science section).

MANUSCRIPTS

[1] *Universality of first-order methods on random and deterministic matrices*, with Nicola Gorini, Tim Kunisky and Chris Jones, 2026.

[2] *Agnostic learning in (almost) optimal time via Gaussian surface area*, with Lucas Slot and Manuel Wiedmer, 2026.

PUBLICATIONS

[3] *Fourier analysis of iterative algorithms*, with Chris Jones, 2025.
Proceedings of the International Colloquium on Automata, Languages and Programming (ICALP).

[4] *New SDP roundings and certifiable approximation for cubic optimization*, with Jun-Ting Hsieh, Pravesh K. Kothari, and Luca Trevisan, 2024.
Proceedings of the Annual ACM-SIAM Symposium on Discrete Algorithms (SODA).

[5] *Discrepancy minimization via regularization*, with Adrian Vladu, 2023.
Proceedings of the Annual ACM-SIAM Symposium on Discrete Algorithms (SODA).

TEACHING ASSISTANTSHIP

(ETH Zürich) Algorithms and Data Structures (Head TA)	Fall 2025
(Bocconi University) 20874 – Algorithms for Optimization and Inference	Fall 2023, Fall 2024
(Bocconi University) 20873 – Analysis of Algorithms and Data Structures	Fall 2023, Fall 2024
(Bocconi University) 30540 – Computing Theory and Algorithms	Spring 2022, Spring 2023
(Bocconi University) Algorithms	Spring 2022

SELECTED COMMUNICATIONS

Fourier analysis of iterative algorithms Jul. 2025
International Colloquium on Automata, Languages and Programming (ICALP 2025), Aarhus.

Fourier analysis of random optimization problems Apr. 2025
Johns Hopkins University.

On maximizing polynomials Dec. 2024
Workshop in memory of Luca Trevisan (1971-2024), Bocconi University.

Understanding iterative algorithms with Fourier diagrams Oct. 2024
DACO seminar, ETH Zürich.

Diagram analysis of iterative algorithms Sep. 2024
Roccella Conference on Inference and AI (poster presentation).

Up to the constant: Spencer's theorem and random matrices Feb. 2024
Computer science seminar, University of Pennsylvania.

New SDP roundings and certifiable approximation for cubic optimization Jan. 2024
Symposium on Discrete Algorithms (SODA 2024), Washington DC.

Discrepancy minimization via regularization Jan. 2023
Symposium on Discrete Algorithms (SODA 2023), Florence.

SERVICE

Reviewing for Symposium on Foundations of Computer Science (FOCS), Symposium on Theory of Computing (STOC), Symposium on Discrete Algorithms (SODA), Conference on Learning Theory (COLT), SIAM Journal on Computing (SICOMP), Journal of the ACM (JACM), Mathematical Statistics and Learning (MSL), Random Structures and Algorithms (RSA).

Ph.D. representative 2022 – 2025

Organization of the theoretical computer science reading group 2021 – 2023

EXTRA-ACADEMIC ACHIEVEMENTS

Bronze medals in the ACM-ICPC Southwestern European contest. 2016, 2017

Bronze medal in the International Olympiad in Informatics (IOI, Taipei) 2014

Ranked 1st in the national IOI selection 2013 – 2014